POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6024B | Semester VI | Theory | 4 Credits

Product Design and Development

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6024B as Product Design and Development in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Roorkee’s Product Design And Development course and NPTEL IIT Kanpur’s Product Design and Manufacturing course. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Product safety, claims, intellectual property, market/financial commitments, regulated-product requirements and final release decisions require current standards, authorised testing and competent legal/technical/commercial review.

Verified source note: Verified sources support product lifecycle and design process, value/function analysis, QFD/customer needs, CAD, robust design, DfX, ergonomics, DFMA, material/process selection, costing, quality, environment, creativity, prototyping, simulation, reverse engineering and industrialisation context. The four modules are a study sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Product Design and Development studies the systematic path from opportunity and user need to a manufacturable, testable and supportable product. It integrates product planning, functional/value analysis, concept development, customer translation, materials/process choice, DfX, quality, prototyping and responsible lifecycle thinking.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Engineering graphics, manufacturing processes, materials, basic mechanics, communication/teamwork, elementary cost awareness and responsible design practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the product-development process, life cycle, opportunity recognition and product planning.
2. Apply functional/value analysis and customer-needs translation to product concepts.
3. Explain material/process selection, costing, robust design and DfX/ergonomics considerations.
4. Explain prototyping, validation, quality, environment, innovation and lifecycle/industrialisation considerations.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO ▯▯▯▯▯▯ exam-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. Define, explain, apply ▯▯▯▯▯ action words ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain product life cycle, product policy and a structured product-development process.	Understand
CO2	Apply functional analysis, value-engineering and customer-needs translation to a simple design problem.	Apply
CO3	Explain material/process selection, costing, robust design, ergonomics and design-for-X principles.	Understand
CO4	Explain prototyping, testing/quality, environmental/lifecycle thinking and innovation/industrialisation pathways.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Product Opportunity, Planning and Life Cycle	Need/opportunity identification; product policy and portfolio awareness; product life cycle; design-process stages; planning, stakeholders, requirements and review gates.	Approx. 14 hours	CO1	Module 1
2	Function, Value and Customer-Need Translation	Functional analysis and FAST awareness; value engineering; customer voice and QFD concept; requirement/specification quality; concept generation, comparison and transparent selection.	Approx. 16 hours	CO2	Module 2
3	Embodiment Design, DfX and Manufacturing Readiness	Materials and manufacturing-process selection; product costing; robust design; DFM, DFA, maintenance and environment; ergonomics, safety and assembly/service awareness.	Approx. 16 hours	CO3	Module 3
4	Prototyping, Validation and Responsible Innovation	Rapid prototyping and simulation; quality planning and testing; design for environment/lifecycle; creativity/frugal innovation; reverse engineering awareness; industrialisation and learning loops.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Product Opportunity, Planning and Life Cycle

Need/opportunity identification; product policy and portfolio awareness; product life cycle; design-process stages; planning, stakeholders, requirements and review gates.

CO1 · Approx. 14 hours

Function, Value and Customer-Need Translation

Functional analysis and FAST awareness; value engineering; customer voice and QFD concept; requirement/specification quality; concept generation, comparison and transparent selection.

CO2 · Approx. 16 hours

Embodiment Design, DfX and Manufacturing Readiness

Materials and manufacturing-process selection; product costing; robust design; DFM, DFA, maintenance and environment; ergonomics, safety and assembly/service awareness.

CO3 · Approx. 16 hours

Prototyping, Validation and Responsible Innovation

Rapid prototyping and simulation; quality planning and testing; design for environment/lifecycle; creativity/frugal innovation; reverse engineering awareness; industrialisation and learning loops.

CO4 · Approx. 14 hours

MODULE 1

Product Opportunity, Planning and Life Cycle

Need/opportunity identification; product policy and portfolio awareness; product life cycle; design-process stages; planning, stakeholders, requirements and review gates.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

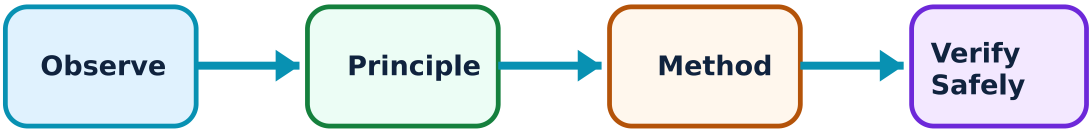
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Product Opportunity, Planning and Life Cycle: identify the system, state its principle, follow the correct method and verify the result safely.

1. From opportunity to product brief−

Product development begins by understanding a problem or opportunity, intended users, context, constraints and organisational goals. A disciplined brief aligns customer value, technical feasibility, cost/time expectations, compliance and lifecycle considerations before detailed design begins.

Study and application method

Create a one-page product brief that separates observed need, target user, use context, measurable goals, constraints, assumptions, stakeholders and questions requiring evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a one-page product brief that separates observed need, target user, use context, measurable goals, constraints, assumptions, stakeholders and questions requiring evidence.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Avoid treating an initial idea as validated demand. Safety, accessibility, legal, environmental and economic assumptions must be checked with appropriate evidence.

2. Product life cycle and development stages−

A product moves through planning, concept development, system/detail design, prototyping, testing, launch, support, retirement and recovery. Decisions made early influence cost, quality, manufacturability, service and environmental outcomes later.

Study and application method

Map a lifecycle for a simple product and identify decisions, evidence and responsible roles at each stage, including feedback paths from use/service to redesign.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

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Common mistake / precaution: Lifecycle maps simplify real projects. Regulated products, high-risk systems and public claims need formal reviews and documented approvals.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Function, Value and Customer-Need Translation

Functional analysis and FAST awareness; value engineering; customer voice and QFD concept; requirement/specification quality; concept generation, comparison and transparent selection.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

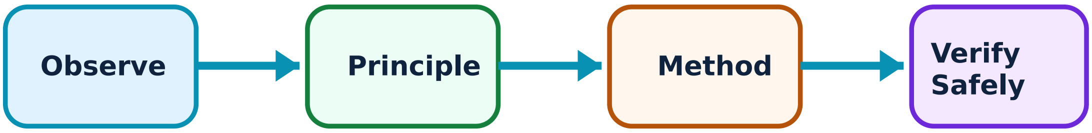
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Function, Value and Customer-Need Translation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Function and value analysis−

Value engineering examines function relative to required performance and resource use. Functional analysis describes what a product must do before deciding how it will do it, which can expose unnecessary complexity or missed alternatives.

Study and application method

Write verb–noun function statements for a product, organise basic/supporting functions in a simple diagram and identify one candidate improvement while preserving essential requirements.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write verb–noun function statements for a product, organise basic/supporting functions in a simple diagram and identify one candidate improvement while preserving essential requirements.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Cost reduction is not value improvement if safety, reliability, user experience, legality or lifecycle cost is degraded. Record trade-offs explicitly.

2. Customer needs, QFD and concept selection−

Customer needs must be translated into measurable engineering characteristics. QFD and similar matrices make relationships, priorities and technical conflicts visible; concept selection compares alternatives against defined criteria and evidence.

Study and application method

Build a small needs-to-metrics table with user need, measurable requirement, unit/target, evidence source, priority and possible trade-off. Use a weighted concept comparison with stated assumptions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build a small needs-to-metrics table with user need, measurable requirement, unit/target, evidence source, priority and possible trade-off. Use a weighted concept comparison with stated assumptions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Weights and scores can create false precision. Include uncertainty, user diversity, safety constraints and reasons a concept may still require testing.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Embodiment Design, DfX and Manufacturing Readiness

Materials and manufacturing-process selection; product costing; robust design; DFM, DFA, maintenance and environment; ergonomics, safety and assembly/service awareness.

Approx. 16 hours **CO3** **Understand · Apply**

Learning goals

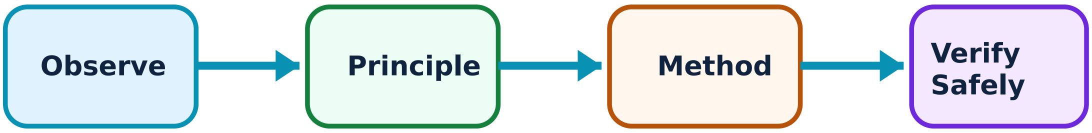
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Embodiment Design, DfX and Manufacturing Readiness: identify the system, state its principle, follow the correct method and verify the result safely.

1. Material, process and cost selection−

Material and process selection match functional requirements with properties, geometry, production volume, quality, supply, cost and environmental conditions. Product cost includes more than material: tooling, processing, assembly, quality, logistics, service and end-of-life can matter.

Study and application method

Create a selection table for a component with required function, loads/environment, candidate materials/processes, evidence source, cost drivers, risk and data still needed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a selection table for a component with required function, loads/environment, candidate materials/processes, evidence source, cost drivers, risk and data still needed.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not approve materials or processes from a generic comparison table. Confirm standards, suppliers, testing, compatibility, safety and actual production conditions.

2. Robust design, ergonomics and design for X−

Robust design seeks acceptable performance despite variation. DfX includes design for manufacture, assembly, maintenance, environment and other lifecycle needs. Ergonomics considers human capabilities, limits, interaction, usability and error reduction.

Study and application method

Perform a DfX review of a simple product: part count, joining, tolerance sensitivity, access for assembly/service, user interaction, material separation and likely waste points.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Perform a DfX review of a simple product: part count, joining, tolerance sensitivity, access for assembly/service, user interaction, material separation and likely waste points.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use a generic ergonomic or safety checklist as proof of compliance. Representative-user evaluation and relevant standards/testing are necessary.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Prototyping, Validation and Responsible Innovation

Rapid prototyping and simulation; quality planning and testing; design for environment/lifecycle; creativity/frugal innovation; reverse engineering awareness; industrialisation and learning loops.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

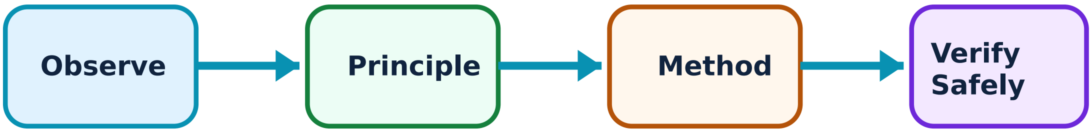
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Prototyping, Validation and Responsible Innovation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Prototypes, simulation and validation−

Prototypes range from sketches and mock-ups to functional models and production-intent units. Simulation and testing reduce uncertainty when they represent the intended use and are linked to clear acceptance criteria. Validation asks whether the solution meets real user and system needs.

Study and application method

Plan a prototype test with learning question, fidelity, scenario, measurement/observation, acceptance criterion, safety boundary, sample/participant limitation and next decision.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Plan a prototype test with learning question, fidelity, scenario, measurement/observation, acceptance criterion, safety boundary, sample/participant limitation and next decision.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A prototype that works once is not proof of safety, reliability or production readiness. Testing must match risk, use conditions and applicable requirements.

2. Quality, environment, innovation and release−

Quality planning links requirements, controls and feedback. Design for environment considers materials, energy, durability, repair, recovery and end-of-life. Creativity and frugal innovation broaden options, while industrialisation requires controlled design data, process capability and ongoing improvement.

Study and application method

Prepare a product-release evidence map: requirement, verification method, result owner, residual risk, lifecycle/environment consideration, revision status and approval/escalation point.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a product-release evidence map: requirement, verification method, result owner, residual risk, lifecycle/environment consideration, revision status and approval/escalation point.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not make performance, sustainability or market claims without substantiation. Patent and freedom-to-operate questions require qualified intellectual-property guidance.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Product Opportunity, Planning and Life Cycle

Use the concept flow to revise observation, principle, method and verification.

Module 2: Function, Value and Customer-Need Translation

Use the concept flow to revise observation, principle, method and verification.

Module 3: Embodiment Design, DfX and Manufacturing Readiness

Use the concept flow to revise observation, principle, method and verification.

Module 4: Prototyping, Validation and Responsible Innovation

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Write a product brief and lifecycle map for a locally relevant everyday product.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Perform a functional/value analysis and needs-to-metrics translation for a chosen design challenge.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Use a transparent concept-selection matrix that lists criteria, assumptions, uncertainty and next validation action.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete a DfX review covering manufacture, assembly, maintenance, ergonomics and environmental recovery.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Plan an ethical, safe prototype-validation exercise with measurable learning objectives and defined limitations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Product Opportunity, Planning and Life Cycle

Explain From opportunity to product brief and state one correct method, application or precaution.−

Product development begins by understanding a problem or opportunity, intended users, context, constraints and organisational goals. A disciplined brief aligns customer value, technical feasibility, cost/time expectations, compliance and lifecycle considerations before detailed design begins.

Answer framework: Create a one-page product brief that separates observed need, target user, use context, measurable goals, constraints, assumptions, stakeholders and questions requiring evidence.

Important point: Avoid treating an initial idea as validated demand. Safety, accessibility, legal, environmental and economic assumptions must be checked with appropriate evidence.

Explain Product life cycle and development stages and state one correct method, application or precaution.−

A product moves through planning, concept development, system/detail design, prototyping, testing, launch, support, retirement and recovery. Decisions made early influence cost, quality, manufacturability, service and environmental outcomes later.

Answer framework: Map a lifecycle for a simple product and identify decisions, evidence and responsible roles at each stage, including feedback paths from use/service to redesign.

Important point: Lifecycle maps simplify real projects. Regulated products, high-risk systems and public claims need formal reviews and documented approvals.

Module 2 — Function, Value and Customer-Need Translation

Explain Function and value analysis and state one correct method, application or precaution.−

Value engineering examines function relative to required performance and resource use. Functional analysis describes what a product must do before deciding how it will do it, which can expose unnecessary complexity or missed alternatives.

Answer framework: Write verb–noun function statements for a product, organise basic/supporting functions in a simple diagram and identify one candidate improvement while preserving essential requirements.

Important point: Cost reduction is not value improvement if safety, reliability, user experience, legality or lifecycle cost is degraded. Record trade-offs explicitly.

Explain Customer needs, QFD and concept selection and state one correct method, application or precaution.−

Customer needs must be translated into measurable engineering characteristics. QFD and similar matrices make relationships, priorities and technical conflicts visible; concept selection compares alternatives against defined criteria and evidence.

Answer framework: Build a small needs-to-metrics table with user need, measurable requirement, unit/target, evidence source, priority and possible trade-off. Use a weighted concept comparison with stated assumptions.

Important point: Weights and scores can create false precision. Include uncertainty, user diversity, safety constraints and reasons a concept may still require testing.

Module 3 — Embodiment Design, DfX and Manufacturing Readiness

Explain Material, process and cost selection and state one correct method, application or precaution.−

Material and process selection match functional requirements with properties, geometry, production volume, quality, supply, cost and environmental conditions. Product cost includes more than material: tooling, processing, assembly, quality, logistics, service and end-of-life can matter.

Answer framework: Create a selection table for a component with required function, loads/environment, candidate materials/processes, evidence source, cost drivers, risk and data still needed.

Important point: Do not approve materials or processes from a generic comparison table. Confirm standards, suppliers, testing, compatibility, safety and actual production conditions.

Explain Robust design, ergonomics and design for X and state one correct method, application or precaution.−

Robust design seeks acceptable performance despite variation. DfX includes design for manufacture, assembly, maintenance, environment and other lifecycle needs. Ergonomics considers human capabilities, limits, interaction, usability and error reduction.

Answer framework: Perform a DfX review of a simple product: part count, joining, tolerance sensitivity, access for assembly/service, user interaction, material separation and likely waste points.

Important point: Do not use a generic ergonomic or safety checklist as proof of compliance. Representative-user evaluation and relevant standards/testing are necessary.

Module 4 — Prototyping, Validation and Responsible Innovation

Explain Prototypes, simulation and validation and state one correct method, application or precaution.−

Prototypes range from sketches and mock-ups to functional models and production-intent units. Simulation and testing reduce uncertainty when they represent the intended use and are linked to clear acceptance criteria. Validation asks whether the solution meets real user and system needs.

Answer framework: Plan a prototype test with learning question, fidelity, scenario, measurement/observation, acceptance criterion, safety boundary, sample/participant limitation and next decision.

Important point: A prototype that works once is not proof of safety, reliability or production readiness. Testing must match risk, use conditions and applicable requirements.

Explain Quality, environment, innovation and release and state one correct method, application or precaution.−

Quality planning links requirements, controls and feedback. Design for environment considers materials, energy, durability, repair, recovery and end-of-life. Creativity and frugal innovation broaden options, while industrialisation requires controlled design data, process capability and ongoing improvement.

Answer framework: Prepare a product-release evidence map: requirement, verification method, result owner, residual risk, lifecycle/environment consideration, revision status and approval/escalation point.

Important point: Do not make performance, sustainability or market claims without substantiation. Patent and freedom-to-operate questions require qualified intellectual-property guidance.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Product Opportunity, Planning and Life Cycle.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Function, Value and Customer-Need Translation.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Embodiment Design, DfX and Manufacturing Readiness.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Prototyping, Validation and Responsible Innovation.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Need/opportunity identification; product policy and portfolio awareness; product life cycle; design-process stages; planning, stakeholders, requirements and review gates..
2. Develop a complete answer or practical plan for: Functional analysis and FAST awareness; value engineering; customer voice and QFD concept; requirement/specification quality; concept generation, comparison and transparent selection..
3. Develop a complete answer or practical plan for: Materials and manufacturing-process selection; product costing; robust design; DFM, DFA, maintenance and environment; ergonomics, safety and assembly/service awareness..
4. Develop a complete answer or practical plan for: Rapid prototyping and simulation; quality planning and testing; design for environment/lifecycle; creativity/frugal innovation; reverse engineering awareness; industrialisation and learning loops..

LAST-MILE REVISION

Quick Revision

Module 1: Product Opportunity, Planning and Life Cycle

Need/opportunity identification; product policy and portfolio awareness; product life cycle; design-process stages; planning, stakeholders, requirements and review gates.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Function, Value and Customer-Need Translation

Functional analysis and FAST awareness; value engineering; customer voice and QFD concept; requirement/specification quality; concept generation, comparison and transparent selection.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Embodiment Design, DfX and Manufacturing Readiness

Materials and manufacturing-process selection; product costing; robust design; DFM, DFA, maintenance and environment; ergonomics, safety and assembly/service awareness.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Prototyping, Validation and Responsible Innovation

Rapid prototyping and simulation; quality planning and testing; design for environment/lifecycle; creativity/frugal innovation; reverse engineering awareness; industrialisation and learning loops.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
From opportunity to product brief	Product development begins by understanding a problem or opportunity, intended users, context, constraints and organisational goals.
Product life cycle and development stages	A product moves through planning, concept development, system/detail design, prototyping, testing, launch, support, retirement and recovery.
Function and value analysis	Value engineering examines function relative to required performance and resource use.
Customer needs, QFD and concept selection	Customer needs must be translated into measurable engineering characteristics.
Material, process and cost selection	Material and process selection match functional requirements with properties, geometry, production volume, quality, supply, cost and environmental conditions.
Robust design, ergonomics and design for X	Robust design seeks acceptable performance despite variation.
Prototypes, simulation and validation	Prototypes range from sketches and mock-ups to functional models and production-intent units.
Quality, environment, innovation and release	Quality planning links requirements, controls and feedback.

LEARNING RESOURCES

References

Recommended product-design-and-development references

1. K. T. Ulrich and S. D. Eppinger, Product Design and Development.
2. G. Pahl et al., Engineering Design: A Systematic Approach.
3. G. E. Dieter and L. C. Schmidt, Engineering Design.
4. G. Taguchi et al., Robust Engineering.

Approved public product-design-and-development sources

- https://nptel.ac.in/courses/112107217
- https://nptel.ac.in/courses/112104230

These sources provide the verified study basis while official Course 6024B detail is unavailable; they do not replace product standards, validation protocols, intellectual-property advice, market evidence or authorised release procedures.